



DEALER BEST PRACTICE SERIES

Tool Calibration Lab

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

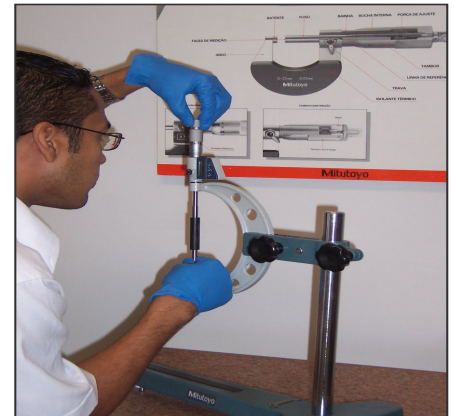
Proper quality control requires that precision measurement tooling be well maintained and go through periodic calibration checks to ensure accuracy. These precision tools include calipers, micrometers, torque wrenches, pressure gauges, and many other tools. Additionally, certification programs such as ISO 9000 require a complete tool calibration program and documentation.

Typically, the tools are sent out to a supplier who has the specialized equipment to perform calibration checks and adjustments. This means that for the CRC must either have spare tools or do without them for the time required to complete the calibration.

An alternative to outsourced tool calibration is to acquire the equipment to perform tool calibrations and do the work in-house. Some dealers also sell this calibration service to customers and other industries as an additional source of revenue.

2.0 Best Practice Description

One dealer began their own internal tool calibration operation with a 20 square-meter facility and two technicians. An initial equipment cost of \$12,000 provided capability to calibrate length, pressure, and torque-measuring devices. Later expanded to 150 square-meters, the lab now employs seven people (one coordinator, four technicians, one apprentice, and one assistant) and serves 120 customers in the automobile, mining, and metallurgy industries.



3.0 Implementation Steps

Analyze the business opportunity.

- Current cost of tooling calibration
- Pricing charged by outside calibration services.
- Potential customers of the service.
- Time lost or spare tool costs.
- Equipment costs
- Staff requirements
- Training.
- Return on investment calculation.

Item	Cost (USD)
Building	\$19,000
Air conditioning, etc.	\$11,400
Calibration tools (torque, temperature, pressure, length, mass)	\$79,400
Furniture	\$7,000
Multifunctional equipment and phones	
Total	\$116,800

Tooling requirements:

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Dimensional Lab

Micrometers
Analog and Digital Dial Indicators
Depth Gauges
Calibers

**Torque Lab**

Torque wrenches
Torque multipliers
Torque drives

**Pressure Lab**

Vacuum and pressure gauges
Valves and pressure switches
Transmitters and transducers
Differential pressure gauges



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Temperature Lab

Thermometers
 Temperature controllers
 Thermocouples
 Transmitters

**Electrical Lab**

Current and Voltage
 Tachometer
 Test Bench Gauges

**4.0 Benefits**

Having an in-house calibration lab allows the CRC and other dealer-owned shops to use a highly specific and tightly controlled tools program. The lab also provides measuring tool training (proper use and maintenance) and professional calibration services to outside customers. Commercial benefits from service sales are significant. In 2006, 4651 tools were calibrated, with 1140 from external customers.

Dealer is ISO 17025 certified to calibrate dimension, torque, temperature, and pressure tooling.

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5.0 Resources Required

Total cost of investments: Approximately \$120,000
Staffing costs

6.0 Supporting Attachments / References

- A complete listing of all calibration tools and costs is attached.

Additional photos:



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7.0 Related Best Practices

None

8.0 Acknowledgements

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